

Online Supplement

Distribution-Free Recalibration of Tail Quantile Forecasts under Temporal Dependence

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This supplement contains full proofs of the asymptotic coverage results stated in the main paper (*Distribution-Free Recalibration of Tail Quantile Forecasts under Temporal Dependence*, Pele et al., 2026). Equation, theorem, and assumption numbering is local to this supplement; the main paper's numbering is cross-referenced in prose. Repository: https://github.com/QuantLet/Conformal_Oracle.

1 Notation and Assumptions

Let $\{r_t\}_{t \geq 1}$ denote the return process and $\{\hat{q}_t^{lo}\}_{t \geq 1}$ the sequence of base α -quantile forecasts with $\hat{q}_t^{lo}$ measurable with respect to $\mathcal{F}_{t-1}$. The one-sided nonconformity score is

$$S_t = \hat{q}_t^{lo} - r_t,$$

with marginal cdf F_S (when it exists). For calibration sample $\mathcal{D}_n = (S_1, \dots, S_n)$ the empirical $(1 - \alpha)$ -quantile is $\hat{q}_v = \hat{F}_n^{-1}(1 - \alpha)$ and the population counterpart is $q_{1-\alpha} = F_S^{-1}(1 - \alpha)$.

Assumption 1 (Main assumption; Assumption 1 of the paper). *Let $S_t = \hat{q}_t^{lo} - r_t$ be strictly stationary and β -mixing with coefficients $\{\beta(k)\}_{k \geq 1}$.*

(A1) *The score density f_S is bounded and bounded away from zero in a neighbourhood of $q_{1-\alpha}$.*

(A2) *Geometric β -mixing: $\beta(k) \leq C\rho^k$ for some $\rho < 1$.*

(A3) *The empirical quantile satisfies $|\hat{q}_v - q_{1-\alpha}| = O_p(\sqrt{\log n/n})$.*

(A4) *S_{n+1} is weakly dependent on $\mathcal{D}_n$ in the sense that the empirical cdf satisfies a uniform concentration bound $\sup_s |\hat{F}_n(s) - F_S(s)| = O_p(\sqrt{\log n/n} + \rho_n)$ for some sequence $\rho_n \rightarrow 0$ driven by the mixing decay.*

2 Main Results (restated)

Theorem 1 (Theorem 3.5 of the paper). *Under Assumption 1,*

$$\mathbb{P}(r_t \geq \hat{q}_t^{lo} - \hat{q}_v) \geq 1 - \alpha - \Delta_n,$$

where $\Delta_n = O\left(\sqrt{\frac{\log n}{n}} + \rho_n\right)$, and $\Delta_n = O\left(\sqrt{\frac{\log n}{n}}\right)$ under geometric β -mixing.

Corollary 1 (Corollary 3.6 of the paper). *Under GARCH(1, 1) dynamics with $E[\log(\alpha_1 \varepsilon_t^2 + \beta_1)] < 0$, the remainder simplifies to $\Delta_n = O(\sqrt{\log n/n})$.*

Proposition 1 (Heuristic local coverage bound for the rolling variant; referenced in Section 3.3 of the paper). *Suppose that within each rolling window $[t - w, t - 1]$, the score process $\{S_\tau\}_{\tau=t-w}^{t-1}$ satisfies Assumption 1 locally with mixing coefficients $\beta_w(k) \leq C_w \rho_w^k$. Then*

$$P\left(r_t \geq \hat{q}_t^{lo} - \hat{q}_{V,t}\right) \geq 1 - \alpha - \Delta_w,$$

where $\Delta_w = O(\sqrt{\log w/w})$ and w is the window size.

3 Supporting Lemmas

Lemma 1 (Inheritance of weak dependence by the score process). *If $\{r_t\}$ is strictly stationary and β -mixing with coefficients $\beta_r(k)$, and $\hat{q}_t^{lo}$ is $\mathcal{F}_{t-1}$ -measurable, then the score process $S_t = \hat{q}_t^{lo} - r_t$ is also strictly stationary and β -mixing with coefficients satisfying $\beta_S(k) \leq \beta_r(k - 1)$ for $k \geq 2$.*

Proof. Stationarity. Since $\{r_t\}$ is strictly stationary by assumption, the joint distribution of any finite collection $(r_{t_1}, \dots, r_{t_m})$ is invariant under time shifts. The base forecast $\hat{q}_t^{lo}$ is $\mathcal{F}_{t-1}$ -measurable, meaning it is a measurable function of $(r_1, \dots, r_{t-1})$; write $\hat{q}_t^{lo} = g_t(r_1, \dots, r_{t-1})$. Under the stationarity of $\{r_t\}$, the function g_t applied to the shifted sequence yields the same marginal distribution for each t : the joint distribution of $(r_s, r_{s+1}, \dots)$ does not depend on s , so neither does the distribution of g_t applied to the preceding returns. It follows that $S_t = \hat{q}_t^{lo} - r_t$ has the same marginal distribution for each t . More generally, the joint distribution of $(S_{t_1}, \dots, S_{t_m})$ depends only on the gaps $t_2 - t_1, \dots, t_m - t_{m-1}$, hence $\{S_t\}$ is strictly stationary.

β -mixing. Let $\sigma_t^{-\infty} = \sigma(S_s : s \leq t)$ and $\sigma_t^{+\infty} = \sigma(S_s : s \geq t)$ denote the past and future σ -algebras of the score process. Since S_t is a measurable function of $(r_1, \dots, r_t)$, we have $\sigma_t^{-\infty} \subseteq \sigma(r_s : s \leq t) =: \mathcal{R}_t^-$. Similarly, S_{t+k} depends on $(r_1, \dots, r_{t+k})$, but through $\hat{q}_{t+k}^{lo}$ it depends on $(r_1, \dots, r_{t+k-1})$ and through r_{t+k} on r_{t+k} directly, so $\sigma_{t+k}^{+\infty} \subseteq \sigma(r_s : s \geq t+k-1)$. For the β -mixing coefficient,

$$\beta_S(k) = \sup_t \frac{1}{2} \sup_{i,j} |\mathbb{P}(A_i \cap B_j) - \mathbb{P}(A_i)\mathbb{P}(B_j)|,$$

where the inner supremum is over finite partitions $\{A_i\}$ of $\sigma_t^{-\infty}$ and $\{B_j\}$ of $\sigma_{t+k}^{+\infty}$. Since $\sigma_t^{-\infty} \subseteq \mathcal{R}_t^-$ and $\sigma_{t+k}^{+\infty} \subseteq \sigma(r_s : s \geq t+k-1)$, we obtain

$$\beta_S(k) \leq \beta_r(k-1).$$

In particular, if $\beta_r(k)$ decays geometrically (or satisfies $\sum \beta_r(k)^\eta < \infty$), then so does $\beta_S(k)$. $\square$

Lemma 2 (Empirical quantile deviation under β -mixing). *Under Assumption 1, let $\hat{q}_V = \hat{F}_n^{-1}(1-\alpha)$ denote the empirical $(1-\alpha)$ -quantile of the calibration scores $S_1, \dots, S_n$, and let $q_{1-\alpha} = F_S^{-1}(1-\alpha)$ denote the population quantile. Then*

$$|\hat{q}_V - q_{1-\alpha}| = O_p\left(\frac{1}{f_S(q_{1-\alpha})} \left(\sqrt{\frac{\log n}{n}} + \rho_n\right)\right).$$

Proof. By (A4), the empirical cdf satisfies the uniform concentration bound

$$\sup_{s \in \mathbb{R}} |\hat{F}_n(s) - F_S(s)| \leq \varepsilon_n \quad \text{where} \quad \varepsilon_n = O_p\left(\sqrt{\frac{\log n}{n}} + \rho_n\right). \quad (1)$$

This bound follows from the Glivenko–Cantelli theorem for β -mixing sequences (Yu, 1994; Rio, 2017): the key step is a blocking argument that partitions the sample into approximately independent blocks of length ℓ_n separated by gaps of length g_n , chosen so that $g_n \rightarrow \infty$ slowly enough that the total number of blocks $n/(\ell_n + g_n)$ still grows, while the mixing coefficients $\beta(g_n)$ become negligible. The empirical cdf within each block can be treated as approximately i.i.d., and the cross-block remainder is controlled by $\sum_{k \geq g_n} \beta(k)^\eta$, giving the dependence correction term ρ_n .

Fix $\delta > 0$ small. By (A2), F_S admits a density f_S with $f_S(q_{1-\alpha}) \geq c_0 > 0$. By continuity of f_S , there exists a neighbourhood $(q_{1-\alpha} - \delta, q_{1-\alpha} + \delta)$ in which $f_S(s) \geq c_0/2 > 0$. On this neighbourhood, F_S is strictly increasing with

$$F_S(q_{1-\alpha} + h) - F_S(q_{1-\alpha}) = f_S(q_{1-\alpha}) h + O(h^2)$$

for $|h|$ small.

The empirical quantile $\hat{q}_V$ satisfies $\hat{F}_n(\hat{q}_V) = 1 - \alpha + O(1/n)$ (the $O(1/n)$ accounts for the ceiling in the order statistic index). Using (1),

$$\begin{aligned} |F_S(\hat{q}_V) - (1 - \alpha)| &= |F_S(\hat{q}_V) - \hat{F}_n(\hat{q}_V)| + |\hat{F}_n(\hat{q}_V) - (1 - \alpha)| \\ &\leq \varepsilon_n + O(1/n). \end{aligned}$$

Since $F_S(q_{1-\alpha}) = 1 - \alpha$ and F_S is locally invertible near $q_{1-\alpha}$ with derivative $f_S(q_{1-\alpha}) \geq c_0$, the inverse function theorem gives

$$|\hat{q}_V - q_{1-\alpha}| \leq \frac{1}{f_S(q_{1-\alpha})} (\varepsilon_n + O(1/n)) = O_p\left(\frac{1}{c_0} \left(\sqrt{\frac{\log n}{n}} + \rho_n\right)\right),$$

where the $O(1/n)$ term is absorbed since $1/n = o(\sqrt{\log n/n})$. □

4 Proof of Theorem 1

Proof. The proof proceeds in three steps.

Step 1: Reduction to score control. By definition of the one-sided nonconformity score $S_t =$

$$\hat{q}_t^{lo} - r_t,$$

$$S_{n+1} \leq \hat{q}_v \iff \hat{q}_{n+1}^{lo} - r_{n+1} \leq \hat{q}_v \iff r_{n+1} \geq \hat{q}_{n+1}^{lo} - \hat{q}_v.$$

Hence it suffices to show

$$\mathbb{P}(S_{n+1} \leq \hat{q}_v) \geq 1 - \alpha - \Delta_n. \quad (2)$$

Step 2: Relating the empirical threshold to the population quantile. Let $q_{1-\alpha} = F_S^{-1}(1 - \alpha)$. By Lemma 2,

$$|\hat{q}_v - q_{1-\alpha}| \leq \frac{\varepsilon_n}{c_0} \quad (3)$$

with probability at least $1 - \delta_n$, where $\varepsilon_n = O_p(\sqrt{\log n/n} + \rho_n)$, $\delta_n \rightarrow 0$, and $c_0 = f_S(q_{1-\alpha}) > 0$ is the density lower bound from (A2). In particular, $\hat{q}_v \geq q_{1-\alpha} - \varepsilon_n/c_0$ with high probability.

Step 3: Approximate coverage. Condition on the calibration set $\mathcal{D}_n$:

$$\mathbb{P}(S_{n+1} \leq \hat{q}_v) = \mathbb{E}[\mathbb{P}(S_{n+1} \leq \hat{q}_v \mid \mathcal{D}_n)].$$

Under (A1), $\{S_t\}$ is strictly stationary and β -mixing. For β -mixing processes, the conditional distribution of S_{n+1} given $\mathcal{D}_n$ converges to the unconditional (stationary) distribution F_S as mixing coefficients decay: for any measurable set A ,

$$|\mathbb{P}(S_{n+1} \in A \mid \mathcal{D}_n) - F_S(A)| \leq \beta(1),$$

where $\beta(1)$ is the β -mixing coefficient at lag 1. Since $\beta(1)$ is a constant (absorbed into C below),

$$\mathbb{P}(S_{n+1} \leq \hat{q}_v \mid \mathcal{D}_n) \geq F_S(\hat{q}_v) - \beta(1). \quad (4)$$

On the high-probability event where (3) holds, using $F_S(q_{1-\alpha}) = 1 - \alpha$ and the mean value theorem,

$$\begin{aligned} F_S(\hat{q}_v) &= F_S(q_{1-\alpha}) + f_S(\tilde{q})(\hat{q}_v - q_{1-\alpha}) \\ &\geq (1 - \alpha) - f_S(\tilde{q}) \cdot \frac{\varepsilon_n}{c_0} \\ &\geq (1 - \alpha) - \frac{\|f_S\|_\infty}{c_0} \varepsilon_n, \end{aligned} \quad (5)$$

where $\tilde{q}$ lies between $\hat{q}_v$ and $q_{1-\alpha}$ and $\|f_S\|_\infty$ denotes the supremum of the density in a neighbourhood of $q_{1-\alpha}$ (finite under the regularity conditions).

Combining (4) and (5) and taking expectations,

$$\begin{aligned} \mathbb{P}(S_{n+1} \leq \hat{q}_v) &\geq \mathbb{E}[F_S(\hat{q}_v)] - \beta(1) - \mathbb{P}((3) \text{ fails}) \\ &\geq 1 - \alpha - C \left(\sqrt{\frac{\log n}{n}} + \rho_n \right), \end{aligned}$$

where $C > 0$ depends on c_0 , $\|f_S\|_\infty$, $\beta(1)$, and the constants in the concentration inequality of (A4). Setting $\Delta_n = C(\sqrt{\log n/n} + \rho_n)$ gives (2).

Substituting back the equivalence from Step 1,

$$\mathbb{P}\left(r_{n+1} \geq \hat{q}_{n+1}^{lo} - \hat{q}_v\right) \geq 1 - \alpha - \Delta_n.$$

Finally, since $\rho_n \rightarrow 0$ under the summability condition $\sum_{k=1}^{\infty} \beta(k)^\eta < \infty$ ((A1)), we have $\Delta_n \rightarrow 0$, establishing asymptotic validity. $\square$

5 Proof of Corollary 1

Proof. We verify that all assumptions of Theorem 1 hold with an improved rate.

Strict stationarity and geometric β -mixing of $\{r_t\}$. Under the conditions $\omega > 0$, $\alpha_1, \beta_1 \geq 0$, and the standard top Lyapunov exponent condition $\mathbb{E}[\log(\alpha_1 \varepsilon_t^2 + \beta_1)] < 0$ with $\mathbb{E}[\log^+ |\varepsilon_t|] < \infty$, the GARCH(1,1) process admits a unique strictly stationary solution (Francq and Zakoïan, 2019, Theorem 2.10). Moreover, Francq and Zakoïan (2019, Theorem 3.2) and Lindner (2009) establish that the process $\{(r_t, \sigma_t^2)\}$ is geometrically ergodic and β -mixing with exponentially decaying coefficients: there exist constants $a > 0$ and $\rho \in (0, 1)$ such that

$$\beta_r(k) \leq a \rho^k, \quad k \geq 1. \quad (6)$$

In particular, $\sum_{k=1}^{\infty} \beta_r(k)^\eta < \infty$ for every $\eta \in (0, 1]$, so (A1) is satisfied.

Inheritance by the score process. By Lemma 1, since $\hat{q}_t^{lo}$ is $\mathcal{F}_{t-1}$ -measurable ((A3)), the score process $S_t = \hat{q}_t^{lo} - r_t$ is strictly stationary and β -mixing with $\beta_S(k) \leq \beta_r(k-1) \leq a \rho^{k-1}$, hence also geometrically β -mixing. (A2) and (A4) are inherited under the regularity conditions.

Simplification of the remainder. For geometric mixing with rate (6), the dependence remainder in the empirical process concentration bound takes the form

$$\rho_n = O\left(\sum_{k \geq g_n} \rho^k\right) = O(\rho^{g_n})$$

for any block gap sequence $g_n \rightarrow \infty$. Choosing $g_n = c \log n$ for a suitable constant $c > 1/|\log \rho|$ gives $\rho^{g_n} = O(n^{-c|\log \rho|}) = o(\sqrt{\log n/n})$. Hence the dependence correction is asymptotically negligible and the dominant term becomes

$$\Delta_n = O\left(\sqrt{\frac{\log n}{n}}\right). \quad \square$$

6 Proof of Proposition 1

Proof. Formally applying Theorem 1 to the sub-sample $\{S_\tau\}_{\tau=t-w}^{t-1}$ with $n = w$. The local stationarity assumption ensures that Assumption 1 holds within each window, yielding the same remainder bound with n replaced by w . The result follows window-by-window; no global stationarity is required. This result is heuristic and does not constitute a formal coverage guarantee under general temporal dependence. $\square$

The local stationarity assumption is substantially weaker than requiring global β -mixing over the full sample: it permits regime changes *between* windows provided that within each 250-day block the score process is approximately stationary. For $w = 250$, the bound gives $\Delta_{250} \approx 0.15$, guaranteeing $\geq 84\%$ coverage per window—conservative but non-vacuous. The empirical rolling coverage of 99.1% reported in the main paper substantially exceeds this floor, consistent with the conservativeness observed for the static bound. The bound formalises the direction of the correction under local dependence rather than providing a tight finite-sample guarantee.

7 Sketch for the Joint VaR–ES Conformal Diagnostic

The joint nonconformity score s_t^J introduced in Appendix C of the main paper combines the VaR and ES residuals.

Sketch. Under exchangeability, the rank of s_{n+1}^J is uniform on $\{1, \dots, n+1\}$, giving $P(s_{n+1}^J \leq \hat{q}_J) \geq 1 - \alpha$ by the standard split-conformal argument (Vovk, Gammerman and Shafer, 2005). Under β -mixing, the score $s_t^J = h(r_t, \widehat{\text{VaR}}_t, \widehat{\text{ES}}_t)$ inherits the mixing properties of $\{r_t\}$, and Theorem 1 applies with the same remainder Δ_n . $\square$

References

- Francq, C. and Zakoïan, J.-M. (2019). *GARCH Models: Structure, Statistical Inference and Financial Applications*, 2nd edition, Wiley.
- Lindner, A. (2009). Stationarity, mixing, distributional properties and moments of GARCH(p, q) processes. In *Handbook of Financial Time Series*, T. G. Andersen et al. (eds.), Springer, 43–69.
- Rio, E. (2017). *Asymptotic Theory of Weakly Dependent Random Processes*. Springer.
- Vovk, V., Gammerman, A. and Shafer, G. (2005). *Algorithmic Learning in a Random World*. Springer.
- Yu, B. (1994). Rates of convergence for empirical processes of stationary mixing sequences. *Annals of Probability* 22(1), 94–116.